# CSCI 620 SQL Database Design

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# Original Design Plan

Originally, the database was structured around the following five tables.

1. **Actor**(ActorCode (PK), ActorName, ActorCountryCode, ActorKnownGroupCode, ActorEthnicCode, ActorReligionCode, ActorTypeCode)
2. **EventAction**(EventCode (PK), IsRootEvent, EventBaseCode, EventRootCode, Quadclass, GoldsteinScale, NumMentions, NumSources, NumArticles, AvgTone, CAMEOCodeDescription)
3. **ActorGeo**(GeoActorCode (PK, FK on ActorCode in Actor Table), ActorGeo_Type, ActorGeo_FullName, ActorGeo_CountryCode, ActorGeo_ADMCode, ActorGeo_Lat, ActorGeo_Long, ActorGeo_FeatureId)
4. **ActionGeo**(GeoActionCode (PK, FK on EventCode in EventAction Table), ActionGeo_Type, ActionGeo_FullName, ActionGeo_CountryCode, ActionGeo_ADMCode, ActionGeo_Lat, ActionGeo_Long, ActionGeo_FeatureId)
5. **DataManagement**(GlobalEventId (PK), DateAdded, SourceURL)
6. **ParticipatesIn**(GlobalEventId (PK), Actor1 (FK to ActorCode in Actor Table), Actor2 (FK to ActorCode in Actor Table), EventCode (FK to EventAction Table's EventCode), Day)

However, upon closer analysis, it was determined that these tables were either missing essential information or contained redundant information that might be better stored in a separate table. A discussion of which changes were made is included in the following sections.

## Actor Table

In the current table design, we can identify one primary concern. Since an actor may have multiple religion or type codes, in the table's current form, either they would have to be stored as an array or we would need to include several columns to store each religion and each type code. This would lead to difficulty of access and deletion or lead to many null values, respectively.

In order to solve this, we can remove the ActorReligionCode and ActorTypeCode from the Actor table and instead create two separate tables to store the religion and type codes. We define tables ReligionCode and TypeCode as follows:

1. ReligionCode(ActorCode (PK, FK to ActorCode in Actor Table), ReligionCode)
2. TypeCode(ActorCode (PK, FK to ActorCode in Actor Table), TypeCode)

## EventAction Table

This table can be better organized, since in its current form, it contains a mix of attributes related to the classification of the event and the importance of the event. To improve table design for meaningful queries, we can store these separately by splitting the attributes as follows:

- Structural Information: EventCode, EventBaseCode, EventRootCode, QuadClass, GoldsteinScale, CAMEOCodeDescription
- Event Importance: IsRootEvent, NumMentions, NumSources, NumArticles, AvgTone,

With this in mind, we separate the EventAction Table into the following two tables:

1. EventAction(EventCode (PK), EventBaseCode, EventRootCode, Quadclass, GoldsteinScale, CAMEOCodeDescription)
2. EventImportance(GlobalEventId (PK), IsRootEvent, NumMentions, NumSources, NumArticles, AvgTone, CAMEOCodeDescription)

## ActorGeo Table:

The Actor Geography attributes are defined as an enhancement to the CAMEO taxonomy and specify geographic references for this event in the text. Since this is specific to an event rather than to a type of event, the current table's primary key is not specific enough to identify the event being referenced. As such, we can add the feature GlobalEventId to the table and update the ActorGeo Table's definition to the following.

1. ActorGeo(GlobalEventId (PK), GeoActorCode (FK on ActorCode in Actor Table), ActorGeo_Type, ActorGeo_Fullname, ActorGeo_CountryCode, ActorGeo_ADMCode, ActorGeo_Lat, ActorGeo_Long, ActorGeo_FeatureId)

## ActionGeo Table:

We similarly update the ActionGeo table as follows:

1. **ActionGeo**(GlobalEventId (PK), GeoActionCode (PK, FK on EventCode in EventAction Table), ActionGeo_Type, ActionGeo_Fullname, ActionGeo_CountryCode, ActionGeo_ADMCode, ActionGeo_Lat, ActionGeo_Long, ActionGeo_FeatureId)

With these corrections in mind, we arrive at the following updated database structure:

2. **Actor**(ActorCode (PK), ActorName, ActorCountryCode, ActorKnownGroupCode, ActorEthnicCode)
3. **ReligionCode**(ActorCode (PK, FK to ActorCode in Actor Table), ReligionCode)
4. **TypeCode**(ActorCode (PK, FK to ActorCode in Actor Table), TypeCode)
5. **EventAction**(EventCode (PK), EventBaseCode, EventRootCode, Quadclass, GoldsteinScale, CAMEOCodeDescription)
6. **EventImportance**(GlobalEventId (PK), IsRootEvent, NumMentions, NumSources, NumArticles, AvgTone)
7. **ActorGeo**(GlobalEventId (PK), GeoActorCode (FK on ActorCode in Actor Table), ActorGeo_Type, ActorGeo_FullName, ActorGeo_CountryCode, ActorGeo_ADMCode, ActorGeo_Lat, ActorGeo_Long, ActorGeo_FeatureId)
8. **ActionGeo**(GlobalEventId (PK), GeoActionCode (PK, FK on EventCode in EventAction Table), ActionGeo_Type, ActionGeo_FullName, ActionGeo_CountryCode, ActionGeo_ADMCode, ActionGeo_Lat, ActionGeo_Long, ActionGeo_FeatureId)
9. **DataManagement**(GlobalEventId (PK), DateAdded, SourceURL)
10. **ParticipatesIn**(GlobalEventId (PK), Actor1 (FK to ActorCode in Actor Table), Actor2 (FK to ActorCode in Actor Table), EventCode (FK to EventAction Table's EventCode), Day)

We can now create the tables in SQL and define the types of each field below. For all varchar attributes, we identified the current max length of the varchars and accordingly select a size larger than that maximum for our tables.

| Attribute Name | Type | Corresponding Column in DF |
|---|---|---|
| Actor Table | | |
| ActorCode<br>**PK** | Varchar (30) | ActorCode |
| ActorName | Varchar (100) | Actor1Name, Actor2Name |
| ActorCountryCode | Char (length 3 since it is a 3 character cameo code) | Actor1CountryCode, Actor2CountryCode |
| ActorKnownGroupCode | Varchar (10) | Actor1KnownGroupCode, Actor2KnownGroupCode |
| ActorEthnicCode | Varchar (10) | Actor1EthnicCode, Actor2EthnicCode |
| ReligionCode Table | | |
| ActorCode<br>**PK, FK to ActorCode in Actor Table** | Varchar (30) (taken from Actor table) | ActorCode |
| ReligionCode | Varchar (10) | ReligionCode |
| TypeCode Table | | |
| ActorCode<br>**PK, FK to ActorCode in Actor Table** | Varchar (30) (taken from Actor table) | ActorCode |
| TypeCode | Varchar (10) | TypeCode |
| EventAction Table | | |
| EventCode<br>**PK** | Varchar (10) | EventCode |
| EventBaseCode | Varchar (10) | EventBaseCode |
| EventRootCode | Varchar (10) | EventRootCode |
| QuadClass | Int | QuadClass |
| GoldsteinScale | Decimal | GoldsteinScale |
| CAMEOCodeDescription | Varchar (100) | CAMEOCodeDescription |
| EventImportance Table | | |
| GlobalEventId<br>**PK** | Int | GlobalEventId |
| IsRootEvent | Int (0 or 1 / binary) | IsRootEvent |
| NumMentions | Int | NumMentions |
| NumSources | Int | NumSources |
| NumArticles | Int | NumArticles |
| AvgTone | Decimal | AvgTone |

| ActorGeo Table | | |
|---|---|---|
| GlobalEventId<br>**PK** | Int | GlobalEventId |
| GeoActorCode<br>**FK to ActorCode in Actor Table** | Varchar (30) (taken from Actor Table) | Actor1Code,<br>Actor2Code |
| ActorGeo_Type | Int | Actor1Geo_Type,<br>Actor2Geo_Type |
| ActorGeo_FullName | Varchar (150) | Actor1Geo_Fullname,<br>Actor2Geo_Fullname |
| ActorGeo_CountryCode | Char (2 character code) | Actor1Geo_CountryCode,<br>Actor2Geo_CountryCode |
| ActorGeo_ADMCode | Char (4 character code) | Actor1Geo_ADMCode,<br>Actor2Geo_ADMCode |
| ActorGeo_Lat | Decimal | Actor1Geo_Lat, Actor2Geo_Lat |
| ActorGeo_Long | Decimal | Actor1Geo_Long,<br>Actor2Geo_Long |
| ActorGeo_FeatureId | Int | Actor1Geo_FeatureId,<br>Actor2Geo_FeatureId |
| ActionGeo Table | | |
| GlobalEventId<br>**PK** | Int | GlobalEventId |
| GeoActionCode<br>**FK to EventCode in EventAction Table** | Varchar (10) (taken from EventAction Table) | GeoActionCode |
| ActionGeo_Type | Int | ActionGeo_Type |
| ActionGeo_FullName | Varchar (150) | ActionGeo_Fullname |
| ActionGeo_CountryCode | Char (2 character code) | ActionGeo_CountryCode |
| ActionGeo_ADMCode | Char (4 character code) | ActionGeo_ADMCode |
| ActionGeo_Lat | Decimal | ActionGeo_Lat |
| ActionGeo_Long | Decimal | ActionGeo_Long |
| ActionGeo_FeatureId | Int | ActionGeo_FeatureId |
| DataManagement Table | | |
| GlobalEventId<br>**PK** | Int | GlobalEventId |
| DateAdded | DateTime | DateAdded |
| SourceURL | Text | SourceURL |
| ParticipatesIn Table | | |

| | | |
|---|---|---|
| GlobalEventId<br>**PK** | Int | GlobalEventId |
| Actor1<br>**FK to ActorCode in Actor Table** | Varchar (30) (taken from Actor Table) | Actor1 |
| Actor2<br>**FK to ActorCode in Actor Table** | Varchar (30) (taken from Actor Table) | Actor2 |
| EventCode<br>**FK to EventAction in EventCode Table** | Varchar (10) (taken from EventCode Table) | EventCode |
| Day | DateTime | Day |